

MH1301 Discrete Mathematics

Handout 8: Graph Theory (2): Traversing Graphs

This is the second handout on graph theory and we focus mainly on paths, which is a sequence of vertices in the graph that are connected by edges. We learn how to solve Euler's Königsberg Bridges problem in this handout.

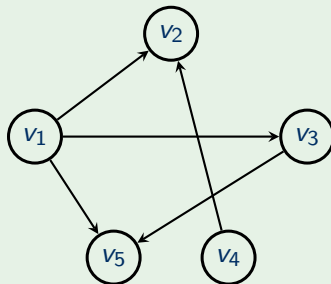
- Graph connectivity
- Euler circuits and Euler paths
- Hamilton circuits and Hamilton paths

Directed Graphs

Recall: A graph $G = (V, E)$ is called **directed** if E contains **ordered** pairs of vertices of V . That is, each edge has a direction.

Example

- $V = \{v_1, v_2, v_3, v_4, v_5\}$
- $E = \{(v_1, v_2), (v_1, v_3), (v_1, v_5), (v_4, v_2), (v_3, v_5)\}$



Degree (Directed Graphs)

Definition 4 (Section 10.2)

For any edge $e = (u, v)$ in a directed graph G , vertex u is called the *initial vertex* of e , and vertex v is called the *terminal vertex* of e .

Definition 5 (Section 10.2)

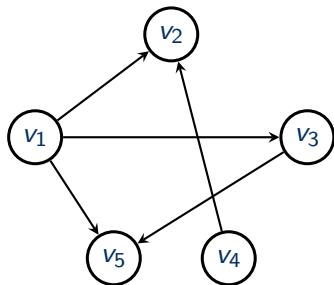
In a directed graph $G = (V, E)$,

- the *in-degree* of a vertex v , denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex.
- the *out-degree* of a vertex v , denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex.

Note: a self-loop at a vertex contributes one to both the in-degree and out-degree of that vertex.

Example

What are the degrees of the vertices?



Solution:

- $\deg^-(v_1) = 0$, $\deg^+(v_1) = 3$,
- $\deg^-(v_2) = 2$, $\deg^+(v_2) = 0$,
- $\deg^-(v_3) = 1$, $\deg^+(v_3) = 1$,
- $\deg^-(v_4) = 0$, $\deg^+(v_4) = 1$,
- $\deg^-(v_5) = 2$, $\deg^+(v_5) = 0$.

Directed “Handshaking Lemma”

Directed Handshaking Lemma (Theorem 3, Section 10.2)

Let $G = (V, E)$ be a directed graph with m edges. Then

$$\sum_{v \in V} \deg^{-}(v) = \sum_{v \in V} \deg^{+}(v) = m.$$

$$2m = \sum_{v \in V} \deg(v) \Rightarrow \text{in handout 7}$$

Proof:

Each edge has exactly one initial vertex and one terminal vertex. □

Compare to Handshaking Lemma (Handout 7): There every edge counts twice in $\sum_v \deg(v)$.

Directed/Undirected, Simple/ Multigraphs

Recall:

Graph Type	Self-Loops Allowed	Multiple Edges Allowed	Directed / Undirected
Simple Graph	No	No	Undirected
Multigraph	Yes	Yes	Undirected
Directed Graph	Yes	Yes	Directed
Directed Multigraph	Yes	Yes	Directed

Directed Simple Graph

No

No

Directed

For the remainder of the handout, the discussion applies to **both** simple graphs and multigraphs.

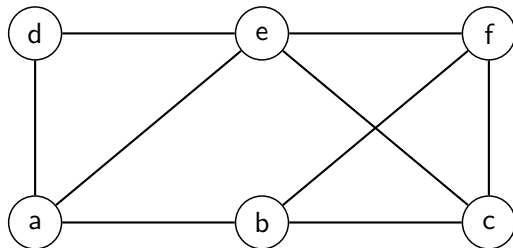
Unless otherwise mentioned, all graphs in this course is simple graph
↳ except for this handout

Paths in Graphs

Definitions 1 & 2 (Section 10.4)

- A **path** in an **undirected** graph G from u to v is a sequence of vertices $(u = w_0, w_1, \dots, w_k, w_{k+1} = v)$, such that $\{w_i, w_{i+1}\}$ are edges in E for all $0 \leq i \leq k$ ("braces" $\{\}$ for undirected edges).
→ vertices don't have to be distinct.
- A **path** in a **directed** graph G from u to v is a sequence of vertices $(u = w_0, w_1, \dots, w_k, w_{k+1} = v)$, such that (w_i, w_{i+1}) are edges in E for all $0 \leq i \leq k$ ("parentheses" $()$ for directed edges).
- Paths can contain the same edge several times.
- The **length** of a path is the number of edges in the path.
- A **simple path** *→ don't have same edge more than once ⇒ cannot repeat.* contains no edge more than once.
 - ▶ A simple path in a directed graph may include **both** the edge (u, v) and the edge (v, u) , but **each at most once**.
- If $u = v$ the path is a **circuit**.

Example: paths



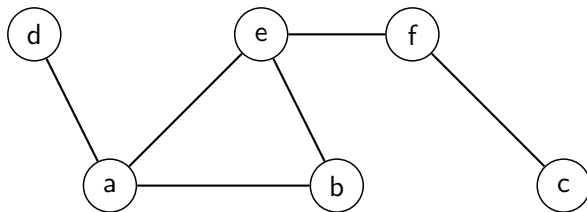
- $d-a-b-c-f$ is a simple path of length 4.
- $d-e-c-b-a-d$ is a circuit of length 5.
- $a-b, e-f$ is not a path, because $(b-e)$ is not an edge. *→ no connection!*
- $d-a-b-c-f-b-a-e$ is a path, but it is not a simple path, because *the edge a-b occurs twice in it*. (length = 7)



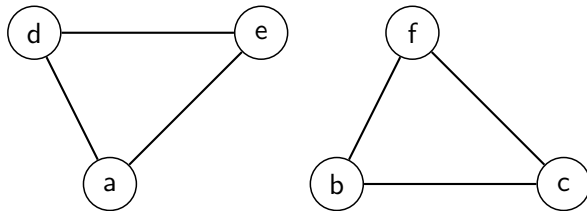
Definition 3 (Section 10.4)

- A graph $G = (V, E)$ (simple or multigraph) is **connected** if for all $u, v \in V$, there is a path between u and v . It is called **disconnected** otherwise.
I doesn't need to be an edge
- A disconnected graph can be partitioned into several connected subgraphs. These are called **connected components**.

Example: connectivity



This graph is connected.



This graph is disconnected. It has 2 connected components.

Connectivity

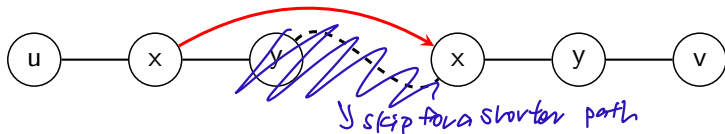
Theorem 1 (Section 10.4)

There is a simple path between every pair of distinct vertices of a connected graph.

Proof: For any two vertices $u, v \in G$, since G is connected, there is at least one path between u and v .

Let P be a path from u to v with the least length. Then P is simple.

- If P is not simple, then there is an edge e being visited at least twice.
- We can find a shorter path from u to v by bypassing the edges in between.



Strongly/Weakly Connected Components

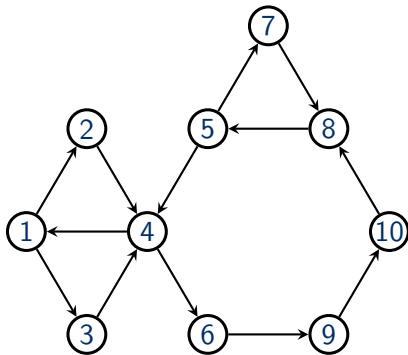
In a directed graph, it is possible that there is a path from u to v but not from v to u .

- Example: a directed path.

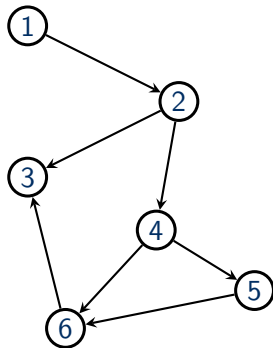
Definition 4 & 5 (Section 10.4)

- A directed graph $G = (V, E)$ is **strongly connected**, if there is a path from u to v as well as a path from v to u , for **all** pairs of vertices u, v .
↳ can go both ways
- A directed graph $G = (V, E)$ is **weakly connected**, if the **underlying undirected graph** is connected. Here, the underlying directed graph is the undirected graph G' that is obtained by removing the directions of all edges of G .

Example

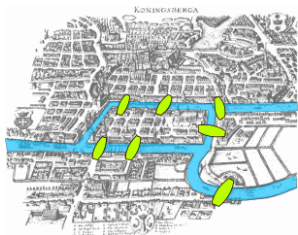


A strongly connected graph.



A weakly connected graph.

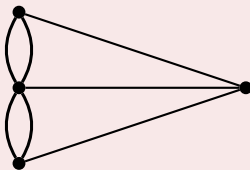
Euler Path/Circuit



- In 1735, Euler wondered if he can have a walk that traverses each of the 7 bridges exactly once.
- His solution is the first use of graph theory.

Königsberg Bridges

Is there a simple path that contains every edge of the following graph?



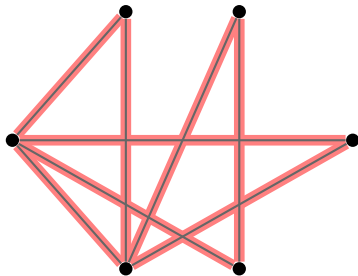
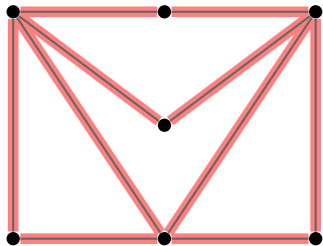
Euler Path/Circuit

all edges are included and not repeated
go back to where we started

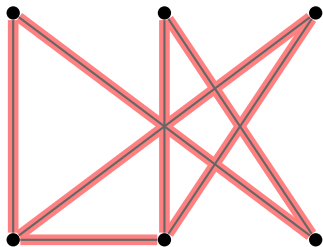
Definition 1 (Section 10.5)

- An **Euler circuit** in a connected graph G is a **simple circuit** containing every edge of G .
→ don't need to go back to where you started.
- An **Euler path** in a connected graph G is a **simple path** containing every edge of G .

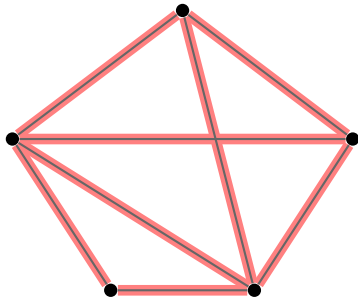
Example: Graphs with an Euler Circuit



Example: Graphs with an Euler Path



↓
not even degree
so not euler's circuit



Question

How do we decide if a graph has an Euler circuit or Euler path?

Theorem 1 (Section 10.5)

A connected graph G with at least two vertices has an Euler circuit if and only if all its vertices have **even** degree.

- It is very easy to test this condition!
- The proof also shows that it is easy to find the Euler circuit.

Euler Circuit: Proof (“Only if”)

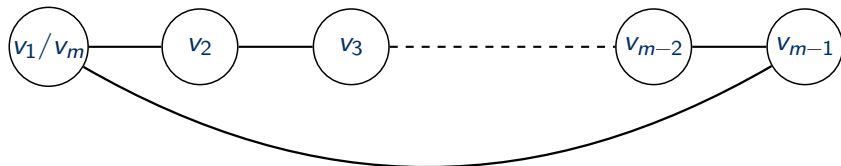
Proof (“only if” direction):

We show that **if** graph G has an Euler circuit, **then** all its vertices have even degree.

Assume G has an Euler circuit $(v_1, v_2, \dots, v_m)$, where $v_m = v_1$. Consider the degree at each vertex v_i .

- For v_1 , edge (v_1, v_2) contributes one to $\deg(v_1)$, edge (v_{m-1}, v_m) contributes one to $\deg(v_m) = \deg(v_1)$.
- For v_i where $(1 < i < m)$, edge (v_{i-1}, v_i) and edge (v_i, v_{i+1}) each contributes one to $\deg(v_i)$.

Hence, for any vertex v , its degree $\deg(v) = \sum_{v_i=v} \deg(v_i)$ is even.



Euler Circuit: Proof (“If”)

Proof (“if” direction):

Now we prove **if** all vertices in graph G have even degree **then** graph G has an Euler circuit.

We first construct a circuit C_1 by the following process:

- Start at an arbitrary vertex v_1 and choose an arbitrary edge (v_1, v_2) incident with v_1 .
- Continue by building a simple path as long as possible until there is no edge to go further. (The path terminates since the graph has a finite number of edges.)

Claim: The resulting path $(v_1, \dots, v_k)$ is a circuit, i.e., $v_1 = v_k$.

Claim: The resulting path $(v_1, \dots, v_k)$ is a circuit, i.e., $v_1 = v_k$.

Proof of Claim:

Let $(v_1, \dots, v_k)$ be the simple path produced by our algorithm and suppose $v_1 \neq v_k$.

Suppose we **remove** the edges of the path from G and consider the degree of v_k in the remaining graph.

- For $1 < i < k$, the degree of v_i is reduced by an even number. (One edge coming in, one edge going out, for each occurrence of v_i on the path.)
- However, for v_k , since $v_1 \neq v_k$, the degree of v_k is reduced by an odd number after removing the edges.

Therefore, there is an unused edge $\{v_k, v_{k+1}\}$.

But the algorithm stopped, so there is no such edge left, which is a contradiction.

Hence $v_1 = v_k$, and this completes the proof of the claim. □

Euler Circuit: Proof (“If”)

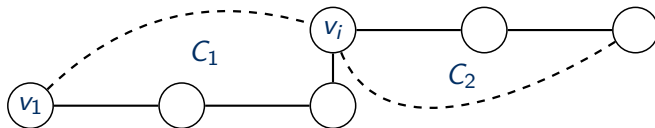
Proof (“if” direction) continue:

From claim, we have constructed a circuit $C_1 = (v_1, \dots, v_k)$.

If this is an Euler circuit we are done. Otherwise,

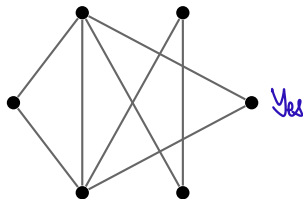
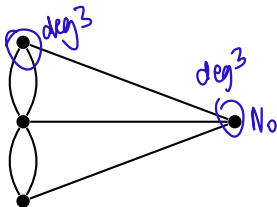
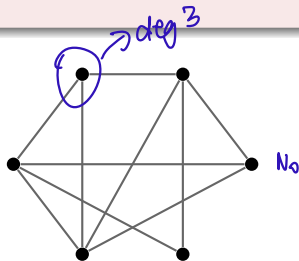
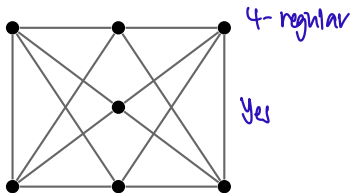
- 1: Remove all edges of C_1 from G . **All vertices still have even degree** in the remaining graph.
- 2: Find a vertex v_i on C_1 with an edge not used in C_1 and construct a new circuit C_2 starting from v_i in the remaining graph.
- 3: Join C_1 and C_2 to form a single new circuit C_1 .

Repeat the Steps 1–3 until it is no longer possible, which is when all edges of G are contained in C_1 and so we have found our Euler circuit.



Example

Do these graphs have an Euler circuit?



Euler Path

Theorem 2 (Section 10.5)

A connected graph G has an Euler path but not an Euler circuit if and only if it has exactly 2 vertices of odd degree.

Proof (“only if” direction):

Suppose G is a connected graph with an Euler path, but no Euler circuit. Let the Euler path be $P = (v_1, \dots, v_m)$, with $v_1 \neq v_m$.

- P leaves v_1 one more time than it enters v_1 .
- P enters v_m one more time than it leaves v_m .
- For all other vertices P leaves as often as it enters.

Therefore, v_1 and v_m have odd degree, all other vertices even degree. $\square$

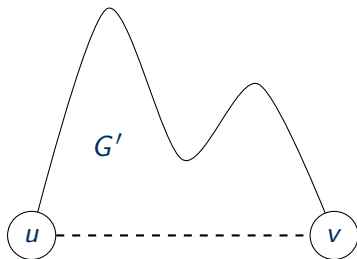
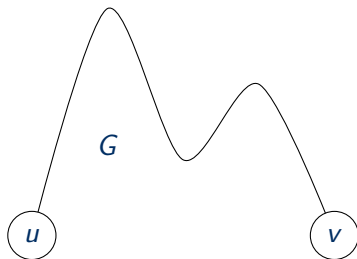


Euler Path

Proof (“if” direction):

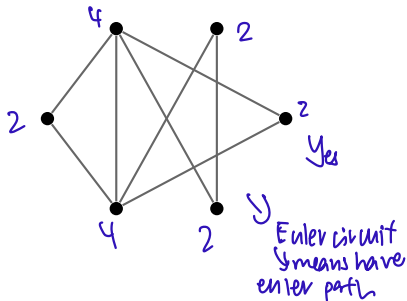
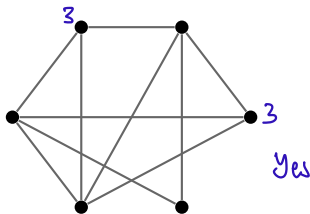
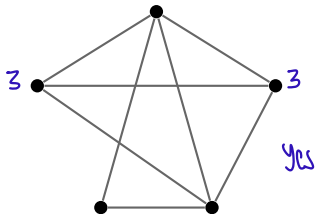
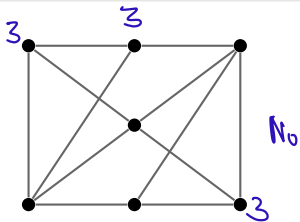
Suppose G has exactly two vertices of odd degree. Let u, v be the vertices of odd degree.

- Add an edge (u, v) to G . Now all vertices have even degree.
- By the previous theorem, this graph has an Euler circuit.
- Removing (u, v) from the circuit will give us an Euler path.



Example

Do these graphs have an Euler path?



Euler Paths/Circuits

- Based on the degrees of the vertices it is easy to decide whether a connected graph has an Euler path/circuit.
- It is also easy to find such an Euler path/circuit.

What if instead of visiting each edge once, we only want to visit each **vertex** once?

Hamilton Path/Circuit

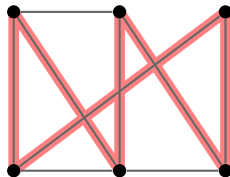
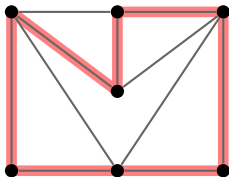
Definition 2 (Section 10.5)

Let $G = (V, E)$ be a graph.

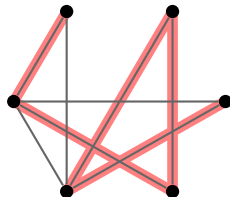
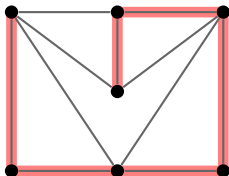
- G has a **Hamilton path**, if there is a path in G that visits every vertex of G exactly once. *→ don't need to go thru on the edges*
- G has a **Hamilton circuit**, if there is a circuit in G that visits every vertex in G exactly once.
I.e., a simple circuit $(x_0, x_1, \dots, x_{n-1}, x_n, x_0)$ is a Hamilton circuit if $(x_0, x_1, \dots, x_{n-1}, x_n)$ is a Hamilton path

Example

Hamilton Circuits



Hamilton Paths



Hamilton Paths/Circuits

Question

How to decide if a graph has a Hamilton Path/Circuit?

- One can “easily” test whether there is a Hamilton path or circuit by trying all $n!$ **possible** Hamilton paths.
 - ▶ Every permutation of the vertices is a candidate for a Hamilton path.
 - ▶ One just needs to check that the corresponding edges are there.
- This is very time-consuming!
- It is unlikely that there is a fast algorithm for these problems.

Reading Assignment

- Rosen: (7th edition)
 - ▶ Chapter 10.4 Connectivity
 - ▶ Chapter 10.5 Euler and Hamilton Paths